

S.No. : 50

NBS 4202/4102

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Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 49902

Roll
No.

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B.Tech. Examination, September 2024

(Special Carry Over Paper)

ENGINEERING PHYSICS

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION - A

1. Attempt all parts of the following : $8 \times 1 = 8$

- (a) Two independent sources could not produce interference, why?
- (b) What do you mean by grating element?
- (c) What is optic axis?
- (d) What are the outcomes of Lave's experiment?
- (e) What is Photon?

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- (f) Write equation of continuity.
- (g) What is the basic principle of optical fibre?
- (h) What are inertial and non-inertial frame of references?

SECTION - B

2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) Calculate the minimum number of lines in a grating which will just resolve the lines of wavelength $5890 \text{ \AA}$ and $5896 \text{ \AA}$ in the second order.
- (b) Calculate De-Broglie wavelength of neutron of energy 12.8 MeV .
- (c) Assuming that all the energy from a 1000 watt lamp is radiated uniformly, calculate the average values of intensities of electric and magnetic fields of radiation at a distance of 2m from the lamp.
- (d) A particle of rest mass m_0 moves with speed $\frac{C}{\sqrt{2}}$. Calculate its mass, momentum and kinetic energy.

SECTION - C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

3. (a) What are Newton's rings? Prove that in reflected light, diameter of the dark ring is proportional to the square root of natural numbers.
- (b) Define resolving power. Derive an expression for the resolving power of grating.
- (c) Describe construction and working of Laurent's half shade polarimeter.
4. (a) Derive time independent Schrodinger wave equation.
- (b) What is Heisenberg uncertainty principle? Apply this to calculate binding energy of electron in an atom.
- (c) What is Bragg's law? Describe Bragg's spectrometer.
5. (a) What is Poynting vector? Derive Poynting theorem.

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- (b) Explain the concept of displacement current. How it led to the modification of Ampere's law?
 - (c) Discuss fibre classification.
6. (a) Derive mass-energy relation. Give some evidence showing its validity.
- (b) Obtain the relativistic formula for addition of velocities. Also show that it is consistent with Einstein's second postulate.
 - (c) Explain why a moving clock appears to go slow to a stationary observer?
